

Incident Investigation: SSH Brute Force & Privilege Escalation on AWS EC2

Role Demonstrated

Tier 1 SOC Analyst / Incident Responder

Scenario

An AWS EC2 instance (ip-172-31-35-28) was targeted by an external attacker. The auth.log file was provided for analysis. The objective was to determine if a breach occurred, identify the attack vector, and document post-compromise activities.

Tools Used

- Splunk Enterprise (local instance)
- Auth.log (Linux authentication log)
- Manual log correlation and timeline analysis

Investigation Summary

Step 1 – Detection

Using Splunk, I identified a high volume of failed SSH authentication attempts from a single source IP: 65.2.161.68. The failures targeted multiple usernames including admin, backup, server_adm, and root within a short time window, indicating an automated brute force attack.

Step 2 – Confirmation of Breach

A successful root login was recorded at 06:31:40 from the same source IP, confirming the attacker gained unauthorized access. A second successful root login occurred at 06:32:44, followed by an extended session.

Step 3 – Post-Compromise Analysis

The attacker performed the following actions after gaining access:

Time	Activity	Evidence
06:34:18	Created a new group (cyberjunkie)	groupadd
06:34:18	Created a new user (cyberjunkie)	useradd
06:35:15	Added user to sudo group	usermod
06:37:57	Read /etc/shadow file	sudo cat /etc/shadow
06:39:38	Downloaded privilege escalation script	curl to GitHub (linper.sh)

Step 4 – Containment Recommendations

- Block source IP 65.2.161.68 at security group level
- Remove malicious user cyberjunkie and associated group
- Revoke and rotate all credentials for root and affected users
- Terminate any active sessions from the source IP
- Audit for persistence mechanisms (cron jobs, SSH keys)

Key Findings

Attack Vector: SSH brute force

Attacker IP: 65.2.161.68

Compromised Users: root, then cyberjunkie (created by attacker)

Indicators of Compromise (IOCs):

- IP: 65.2.161.68
- URL: raw.githubusercontent.com/montysecurity/linper/main/linper.sh
- Usernames: cyberjunkie

Outcome

The incident was successfully detected through log analysis. A full timeline was reconstructed using Splunk queries. Containment steps were documented and prioritized. This investigation demonstrates proficiency in:

- Splunk search and correlation
- Linux authentication log analysis

- Incident documentation and reporting
- Threat hunting for post-exploitation activity

Artifacts

- Full Splunk SPL queries used in investigation
- Timelined attack reconstruction
- IOC list for firewall and EDR blocking

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